

# TSMR: Track-aware Support Manifold Refinement

Frozen inputs



RGB track



Aligned SMPL

+

Metric point cloud

Track ID

Human mask

Neighbor frames

Stage2 trans

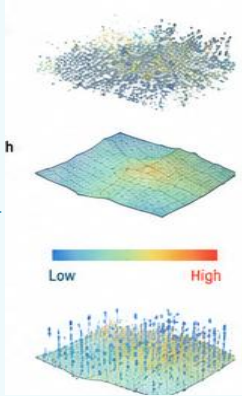
## 1 Build support



Aligned human + scene

Remove human points

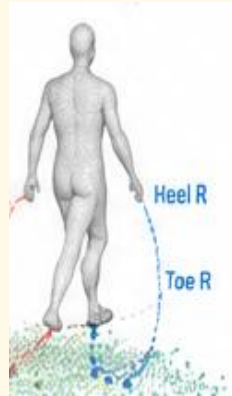
Output: support point, normal, confidence



Local support surface

Fit ground / support

## 2 Propose candidates



Probe feet / body

No-op

Left / right foot

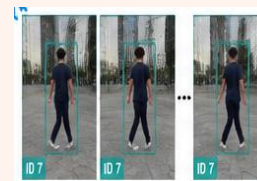
Both feet

Body support

Geometry gives a small candidate set

The network does not invent a large translation

## 3 Select, update, verify

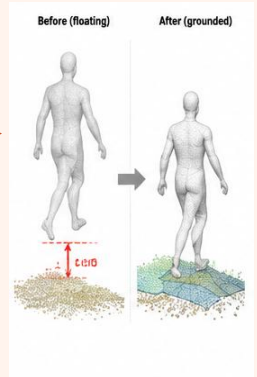


same ID over time

Candidate selector

Choose / abstain

Update translation



Before (floating) After (grounded)

Before -> Grounded

Re-probe

Only translation is changed

Pose, shape, and scene stay fixed

re-probe and repeat x3

TRAIN

candidate selection

translation

clean no-op

temporal consistency

INFERENCE

support mode | refined translation | confidence